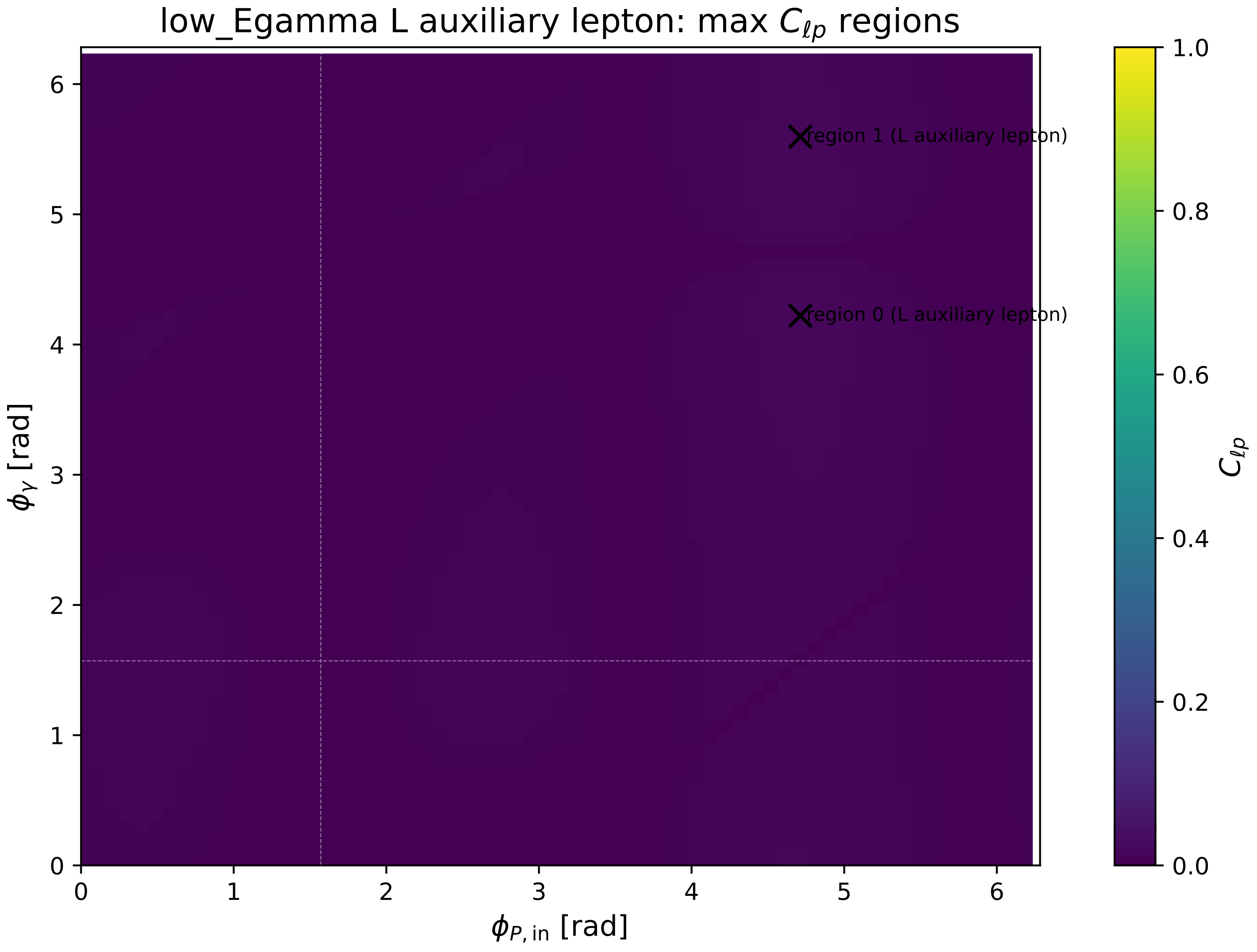
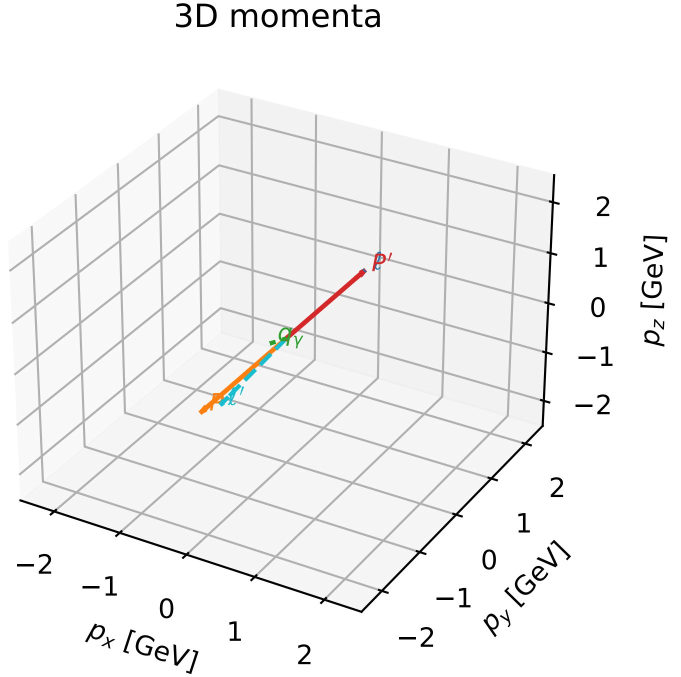


low_Egamma L auxiliary lepton: max $C_{\ell p}$ regions
region 1 (L auxiliary lepton)
region 0 (L auxiliary lepton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\ell p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L auxiliary lepton characteristic max $C_{\ell p}$ configuration

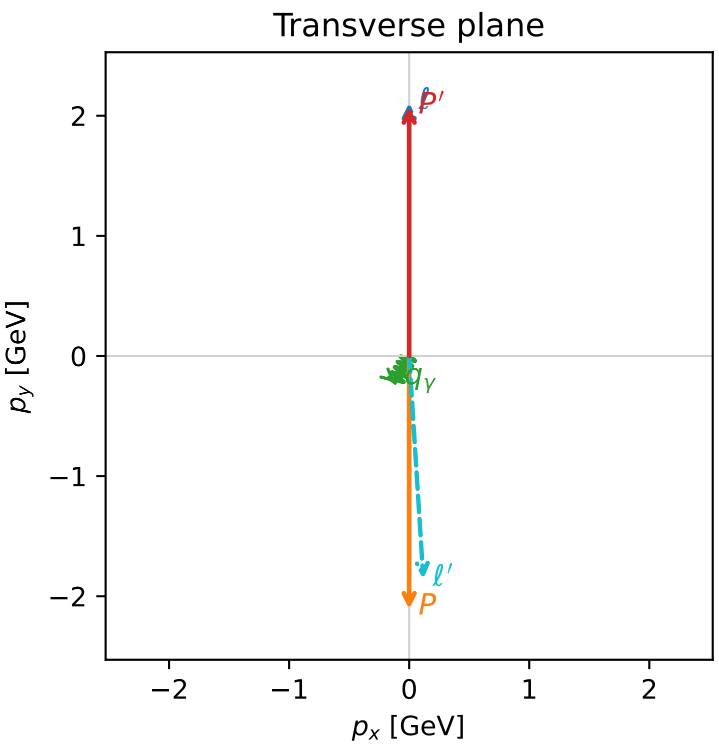


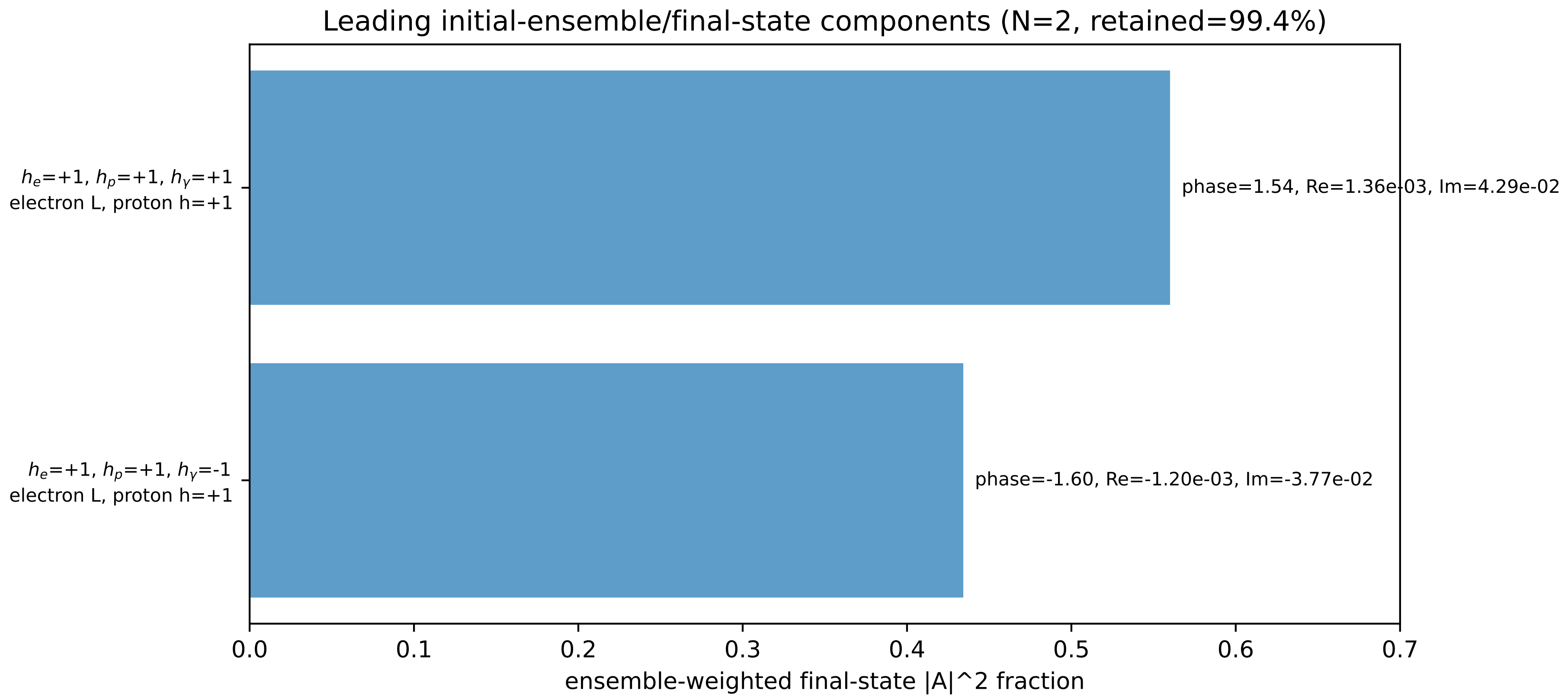

```
c_aux_p_low_egamma_l_lepton_region_0  C_{l_auxp}=0.0171955
region: low_Egamma_L_lepton_region_0  final-pair delta_xy=3.07815 rad
outgoing-state purity: 0.999662
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

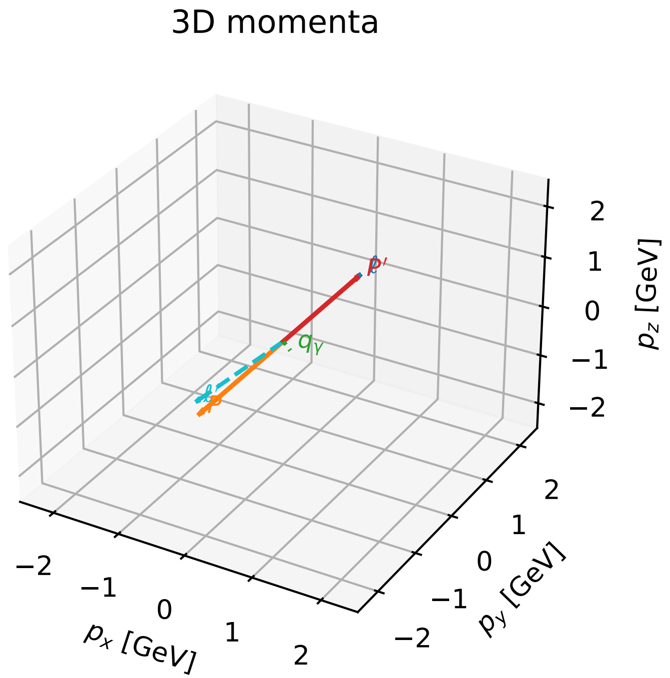
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.07562
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=4.22152
Q^2=15.663, x_B=0.884059, t=-17.493, W^2=2.93397, y=0.902276
$\theta(\ell', \gamma)$=31.7599 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.0000 p_y= 2.1069 p_z= -0.0000
P E= 2.3063 p_x= -0.0000 p_y= -2.1069 p_z= 0.0000
ℓ' E= 2.1108 p_x= 0.1178 p_y= -1.8551 p_z= 0.0000
P' E= 2.2777 p_x= 0.0000 p_y= 2.0756 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1178 p_y= -0.2205 p_z= 0.0000





L auxiliary lepton characteristic max $C_{\ell p}$ configuration




```
c_aux_p_low_egamma_l_lepton_region_1  C_{l_auxp}=0.016886
region: low_Egamma_L_lepton_region_1  final-pair delta_xy=3.03935 rad
outgoing-state purity: 0.999614
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_\ell, h_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.04213
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=5.59596
Q^2=15.9248, x_B=0.8999, t=-17.2113, W^2=2.65122, y=0.901209
$\theta(\ell', \gamma)$=56.4831 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.0000 p_y= 2.1069 p_z= -0.0000
P E= 2.3063 p_x= -0.0000 p_y= -2.1069 p_z= 0.0000
ℓ' E= 2.1413 p_x= -0.1933 p_y= -1.8835 p_z= 0.0000
P' E= 2.2472 p_x= 0.0000 p_y= 2.0421 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1933 p_y= -0.1586 p_z= 0.0000